higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

ELECTRONIC CONTROL AND DIGITAL ELECTRONICS
NQF LEVEL 2

(12041022)

1 March 2019 (X-Paper)
09:00–12:00

Calculators and drawing instruments may be used.

This question paper consists of 6 pages.

NCV

Copyright reserved

TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Start each question on a NEW page.
5. Write neatly and legibly.

NCV

QUESTION 1: COMPONENTS AND CIRCUIT DRAWING

1.1 Explain *covalent bonding* with the aid of a sketch. (4)

1.2 Identify the type of markings for each of the following electronic components:

1.2.1 

1.2.2 

(2 × 1) (2)

1.3 1.3.1 Give the value of the component in QUESTION 1.2.1. (1)

1.3.2 Give the value of the component in QUESTION 1.2.2. (1)

1.4 Explain the function of each of the following components:

1.4.1 Capacitor

1.4.2 Transformer

1.4.3 Relay

(3 × 1) (3)

1.5 Explain, with the aid of a neat labelled sketch, the formation of a PN-junction. (3)

1.6 Draw a labelled circuit diagram of a common emitter amplifier. (6)

[20]

NCV

QUESTION 2: COMPONENTS AND CIRCUIT DRAWING

2.1 Explain *linear integrated circuits* and give ONE example. (2)

2.2 FIGURE 1 shows an LM741.

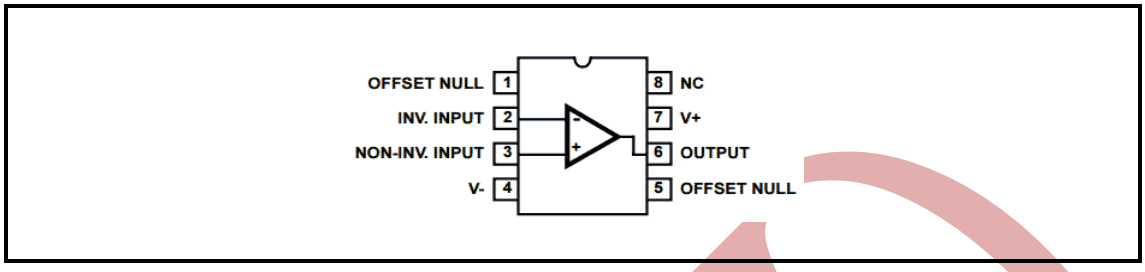


FIGURE 1

Determine the pin number of the output voltage. (1)

2.3 Determine the output voltage in the noninverting mode of an op-amp with input voltage 2 V, feedback resistance 30 kΩ and input resistance 2 kΩ. (3)

2.4

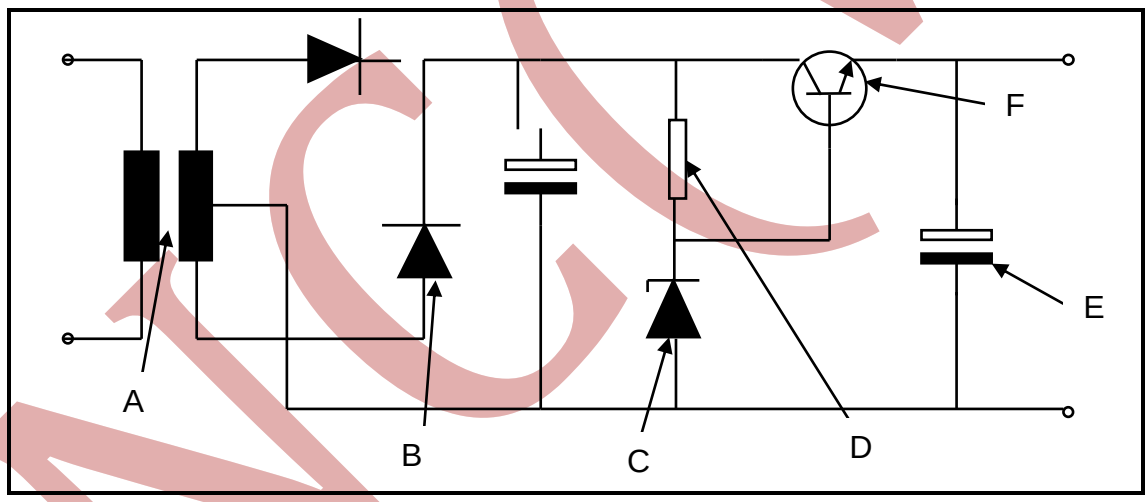


FIGURE 2

Name the IEC symbols in FIGURE 2 by writing only the answer next to the letter (A–F) in the ANSWER BOOK. (6 × 1) (6)

2.5 Explain how a multimeter is used to test a diode. (4)

2.6 Give THREE advantages of using a breadboard while performing an experimental project in the workshop. (3)

2.7 Name THREE safety precautions to take when using solder wire. (3)

2.8 TWO inductors of values 3 mH and 5 mH are connected in parallel.

Calculate the total inductance. (2)

Copyright reserved

2.9 Refer to the table below and determine the maximum current that the LED can handle.

Part no.	Description	Colour	Lens	VF(V) TYP	IF (Ma)
12-001	L30R LED 3 mm	Red	Diffused	1,7	20

(1)
[25]

QUESTION 3: DIGITAL ELECTRONICS

3.1 Give FOUR advantages of digital systems over analogue systems. (4)

3.2 Briefly explain how voltage levels are used to represents bits. (2)

3.3 Distinguish between *analogue signals* and *digital signals*. (2)

3.4 Convert the binary number 101101 to a decimal number. (4)

3.5 Convert the decimal numbers 35_{10} and 14_{10} to binary numbers and add the TWO converted binary numbers. (8)

3.6 Subtract 1100_2 from 1011_2. (2)

3.7 Draw the truth table and describe the operation of the AND gate. (4)

3.8 Draw the IEC symbols for each of the following logic gates:

3.8.1 NAND

3.8.2 NOT

3.8.3 NOR

3.8.4 OR

(4 × 1) (4)

3.9 Define a *flip-flop*. (1)

3.10 Draw the IEC symbol of a D-type flip-flop. (4)
[35]

NCV

QUESTION 4: BASIC COMPUTER COMPONENTS

4.1 Name the main function of computer ports. (2)

4.2 Explain the function of each of the following computer peripherals:

4.2.1 Monitor

4.2.2 Printer

4.2.3 Scanner

(3 × 1) (3)

4.3 Explain how binary levels are stored in the CD memory. (5)
[10]

QUESTION 5: TRANSDUCER

5.1 Name TWO types of electronic probes that can be used to measure the levels of liquids. (2)

5.2 Draw the construction of a photo diode. (4)

5.3 Give TWO advantages of venturi tubes. (2)

5.4 Name TWO types of orifice plates. (2)
[10]

TOTAL: 100

NCVO

Copyright reserved